TEAM DATA SNAPSHOT

What data is ready?

Version liquid-team-data-v1-20261020 / original data preserved

Camera	JPEG stills	MP4s	Exported boxes
vial_fill	1733	5	2676
tank_level	1733	5	12292
cap_seal	1181	5	1846
Total	4647	15	16814

All indexed stills are 640 × 480. The videos have nominal 30 fps. Existing box exports are machine-generated and unreviewed. The source also retains 22,054 per-class hypothesis records, including low-confidence/suppressed entries; those records are not all positive labels.

Accepted detector starter

Profile	Training	Development
Original-image profile	721 distinct source images	157 Run 4 images
Historical profile	1,442 samples: originals + 721 translations	The same 157 images
Legacy liquid targets, originals	354 positive / 367 provisional label-absence	89 positive / 68 provisional label-absence

The starter subset has 551/76/94 originals from Runs 1/2/3 and 157 from Run 4. Ten raw Runs 1–3 vial stills are outside that accepted subset. Across all five runs, 21 vial-fill images lacked a normal label export and require manual review; no empty label files were fabricated for them.

What is not supplied as reviewed truth

No human-approved pool masks, first-visible onset times, confirmed-empty decisions or tracking identities are included. The legacy colored_liquid class can combine stream and pool. It is distinct from the human-review contract's liquid_body and stream classes.

Three unindexed duplicate-name JPEGs are cataloged separately and excluded from the release. Cap/seal has one indexed still with an unassigned run; it remains context-only. Original files and reviewer workspaces were not modified.

FILES & TEAM STARTERS

Download only what you need

Extract archives into one directory; their common top-level folder combines the components

Archive	Contents / audience
00-team-guide.zip	This guide, team architecture handout, inventory, source manifests, contracts and coordinate/split documentation. Everyone starts here.
01-detector-starter.zip	Shared starter images; vial-only YOLO labels/YAMLs; COCO vial, legacy-liquid and two-class box annotations. Teams 1 and 2; Team 3 may reuse the RGB/boxes.
02-vial-source-runs1-4.zip	Original vial-fill JPEGs and videos for Runs 1–4, with matching provisional raw label exports. All teams need sequences for onset/transition work.
03-sam-frames-run4-development.zip	300 lossless Run 4 PNGs, native frame IDs/PTS and machine detector proposals. Development only; no final prompts, temporal recipe, SAM masks or approved masks.
04-other-cameras-runs1-4.zip	Tank-level and cap/seal source context with their camera-local provisional labels. Optional for the current three-team vial/pool task.
05-run5-diagnostic-only.zip	Run 5 images/videos and organizer recipe/timing references, isolated from fitting and model selection. Diagnostic access only.

Formats at a glance

Format	Coordinates / class IDs
YOLO TXT + YAML	Normalized class/center-x/center-y/width/height. Starter vial class = 0. Each YAML resolves within starter/.
COCO detection JSON	Pixel [left,top,width,height]; image file_name relative to starter/. Single-class category = 1. No segmentation masks are implied.
SAM frame manifest	Pixel [left,top,right,bottom] proposals, lossless RGB PNG, exact source frame/native PTS and source/model identities.
CSV / JSON source index	Camera/run, source byte hashes, package paths, split roles and explicit missing/review status.

Use starter/vial-originals.yaml by default. The historical translated profile is separate. For Faster R-CNN, start with starter/coco/legacy-liquid-train-originals.json and legacy-liquid-development.json, using starter/ as the image root. Keep the provisional semantics explicit.

SPLITS & LABEL USE

Preserve the evidence boundaries

Data availability does not make the annotations human-approved

Boundary	Required handling
Training	Runs 1–3 only. Keep every translated image with its original donor. Do not randomly split temporal neighbors or duplicate source content across train/development.
Development	Run 4 has been used for checkpoint selection and iterative inspection. Its 300-frame SAM starter is a development fixture, not a fitting set or blind test.
Diagnostics	Run 5 has already been inspected. Keep it out of fitting, checkpoint choice, threshold tuning and sampling decisions. No untouched blind test is supplied.
Missing liquid target	The image may be a provisional negative for an explicitly declared pilot. It is not a confirmed-empty frame or a reviewed no-pool decision.
Missing raw label file	Unresolved/manual-review case. Never manufacture an empty label file and silently treat it as negative.
Review corrections	Keep original machine labels immutable. Store reviewed decisions separately with source identity, reviewer, status and provenance. Do not silently merge conflicting histories.
Clocks and identities	Still/video frame associations remain unmatched unless reviewed. Organizer timing/recipe files are separate references, not spatial masks, approved pool onsets or inferred physical vial IDs.

What teams can do immediately

Teams 1 and 2 can prototype detectors from the accepted provisional starter while reviewing their target semantics. Team 3 can implement segmentation prompting and evaluate qualitative behavior from the same RGB and detector inputs. Reviewed masks and onset frames are still needed for quantitative segmentation and early-detection claims.

Traceability

metadata/images.csv and images.json identify every indexed source still; videos.json records video hashes and container metadata. starter/samples.json preserves source/run/split and translation ancestry. Each downloadable archive receives a SHA256 checksum and a read-back integrity check. Container frame counts do not claim a full decode of all 15 raw recordings.

Format references: https://docs.ultralytics.com/datasets/detect/ and https://cocodataset.org/#format-data. The package includes data/documentation only; no model weights, executable source, private reviewer workspaces or final solution traces.